

Name: chowutiyu Avilkumar

BRN:- PF51PG25CA044

subject code:- UQ25CA653A

P1 Classify as proposition or not

• (a) $2 + 3 = 5$ - Proposition (true)

• (b) "Close the door" - not a proposition

• (c) $3x + 2 \geq 7$ - not a proposition

• (d) "Bengaluru is in India"

Proposition

P2 truth table $(P \vee q) \rightarrow (-q \wedge q)$

P	q	$P \vee q$	$-P \wedge q$	$(P \vee q) \rightarrow (-P \wedge q)$
1	1	1	0	0
1	0	1	0	0
0	1	1	1	1
0	0	0	0	1

P3 $\neg(P \rightarrow Q) \equiv P \wedge \neg Q$

P	Q	$\neg P$	$\neg Q$	$P \rightarrow Q$	$\neg(P \rightarrow Q)$	$P \wedge \neg Q$
0	0	1	1	1	0	0
0	1	1	0	1	0	0
1	0	0	1	0	1	1
1	1	0	0	1	0	0

P4 Prove $(P \rightarrow Q) \equiv (\neg Q \rightarrow \neg P)$ (contrapositive)
and give a counterexample for $(P \rightarrow Q) \equiv (Q \rightarrow P)$

P	Q	$\neg Q$	$\neg P$	$P \rightarrow Q$	$\neg Q \rightarrow \neg P$
0	0	1	1	1	1
0	1	0	1	1	0
1	0	1	0	0	0
1	1	0	0	1	1

P5 $E(x) = "x \text{ student every assignment}"$
 $P(x) = "x \text{ passed}"$

a) every student who submitted every assignment passed

$$\forall x [E(x) \rightarrow P(x)]$$

b) some student passed without submitting every assignment

$$\exists x [P(x) \wedge \neg E(x)]$$

c) no student both failed and submitted every Assignment

~~Proof A - P(20)~~

$$= \neg [-P(20) \wedge P(60)]$$

s1 $A = \{1, 2, 3, 4\}$

$$B = \{3, 4, 5, 6\}$$

$$A \cup B = \{1, 2, 3, 4, 5, 6\}$$

$$A \cap B = \{3, 4\}$$

$$A - B = \{1, 2\}$$

$$B - A = \{5, 6\}$$

$$U = \{1, 2, 3, 4, 5, 6\}$$

$$C) A' = \{5, 6\} \quad |A'| = 2 \quad |B'| = 2 \quad |A' \cap B'| = 2$$

s2 $(A \cup B)' = A' \cap B'$

$$A = \{a, b, c\}$$

$$B = \{b, c, d\}$$

$$U = \{a, b, c, d, e\}$$

$$A \cup B = \{a, b, c, d\}$$

$$(A \cup B)' = \{e\}$$

$$A' = \{d, e\}$$

$$B' = \{a, e\}$$

$$A' \cap B' = \{e\}$$

Q3. Survey. (Wobets) 1120 students. $|U| = 70$
 (Comics) $|K| = 60$ Comics, $|C \cap K| = 30$
 a) At least one $|C \cup K| = 120$
 b) Exactly one $|C \cup K| - |C \cap K| = 120 - 30 = 90$
 c) Neither $\emptyset$

Q4. $|M| = 128$
 $|P| = 87$
 $|C| = 134$
 $|M \cap P| = 31$
 $|M \cap C| = 54$
 $|P \cap C| = 30$

total failed = 250 $|M \cup P \cup C|$

$$x = |M| + |P| + |C| - (|M \cap P| + |M \cap C| + |P \cap C|) + |M \cap P \cap C|$$

$$250 = 128 + 87 + 134 - (31 + 54 + 30) + x$$

$$250 = 349 - (115) + x$$

$$250 = 234 + x$$

$$250 - 234 = x$$

$$x = 16$$

Q5. complement counting How many 3-digit numbers contain at least one digit 7

→ 252